

JINSUK BAEK

Professor of Computer Science | Winston-Salem State University

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RESEARCH INTERESTS

- Quantum Computing and Quantum Networking
- Intelligent Networked Systems and IoT
- AI-Enabled Robotics and Cyber-Physical Systems

EDUCATION

Ph.D. in Computer Science — University of Houston, TX, USA • *Aug 1999 – Dec 2004*

Dissertation: Scalable Approaches for Tree-Based Reliable Multicast

Advisor: Dr. Jehan-François Pâris

M.Sc. in Computer Science and Engineering — Hankuk University of Foreign Studies, Korea • *Mar 1996 – Feb 1998*

Thesis: An Efficient Parameter Coding Method for Object-Oriented Coder (MPEG-4)

Advisor: Dr. Sangchul Kim

B.Sc. in Computer Science and Engineering — Hankuk University of Foreign Studies, Korea • *Mar 1992 – Feb 1996*

PROFESSIONAL EXPERIENCE

Professor, Department of Computer Science, Winston-Salem State University • *Jul 2016 – Present*

Associate Professor, Department of Computer Science, Winston-Salem State University • *Jul 2010 – Jun 2016*

Assistant Professor, Department of Computer Science, Winston-Salem State University • *Jul 2005 – Jun 2010*

Post-Doctorate Research Associate, Distributed Multimedia Research Group, University of Houston • *Jan 2005 – Jun 2005*

Advisor: Dr. Jehan-François Pâris

Consulting Expert, Rong Gabello Law Firm for Apple, Inc. • *Mar 2005*

Research Assistant, Distributed Multimedia Research Group, University of Houston • *Aug 2002 – Dec 2004*

Advisor: Dr. Jehan-François Pâris

Teaching Assistant, Department of Computer Science, University of Houston • *Aug 2000 – Jul 2002*

Software Developer, IBM Korea (with Leading Soft. Co.), Seoul, Korea • *Jan 1999 – Jun 1999*

Designed and implemented electronic approval systems for an Intranet using Lotus Notes R4.

Research Assistant, Multimedia Information Telecommunication Lab, Hankuk University of Foreign Studies • *Mar 1996 – Feb 1998*

Advisor: Dr. Sangchul Kim

AWARDS & HONORS

- Outstanding Paper Award, 28th IEEE International Conference on Advanced Communication Technology — *Feb 2026*
- Research Award, HBCU Smart City Challenge — *Feb 2023*
- Outstanding Paper Award, 22nd IEEE International Conference on Advanced Communication Technology — *Feb 2020*
- Best Paper Award, 2018 International Conference on Recent Advances in Engineering and Technology — *Oct 2018*
- Outstanding Paper Award, 18th IEEE International Conference on Advanced Communication Technology — *Feb 2016*
- Outstanding Paper Award, 14th IEEE International Conference on Advanced Communication Technology — *Feb 2012*
- Professional Development Travel Grant, Winston-Salem State University — *2006–2019*
- Professional Development Research Grant, Winston-Salem State University — *2017–2019*
- Research Advisor, NC Space Grant Graduate Research Fellowship — *2010–2011*
- Research Advisor, NC Space Grant Graduate Research Fellowship — *2007–2008*
- Research Advisor, NC Space Grant Graduate Research Fellowship — *2006–2007*
- Encouragement Prize, 1st SAMSUNG Electronics Future Technology Research Contest — *2000*
- Infrastructure Studies Student in Japan, Sponsored by SAMSUNG — *1994*

TEACHING EXPERIENCE

Courses Developed and Taught

Undergraduate Level

- Directed Study (CSC 4390) — *Spring 2025, Fall 2026*
- Computer Communication Networks (CSC 4389) — *Spring 2006–2026*
- Hardware and Media Security (CSC 4360) — *Fall 2023*
- Cryptography (CSC 4330) — *Fall 2007–2008, Spring 2016–2019*
- Parallel Computing (CSC 4322) — *Spring 2010, 2012, Fall 2025*
- Analysis of Algorithms (CSC 3331) — *Spring 2025–2026*
- Computer Architecture (CSC 3322) — *Spring 2007, 2009, 2019–2026*
- Operating Systems (CSC 3321) — *Fall 2009–2011, Spring 2026*
- Intro. to Computer Hardware Organization (CSC 2320) — *Fall 2012, 2014–2022, Spring 2020–2022*
- Intro. to Computer Software Systems (CSC 2310) — *Spring 2011, 2013–2016*
- Professional Development Seminar (CSC 2131) — *Fall 2024*
- Intro. to Computer Technology (CSC 1307) — *Fall 2013*
- Computer and Its Use (CSC 1306) — *Fall 2005–2007, 2010–2012, 2014; Summer 2006–2009*

Graduate Level

- Network Security (CST 6309) — *Spring 2006, 2010–2015*
- Network Design, Performance and Evaluation (CST 6308) — *Spring 2008–2009*
- Advanced Computer Architecture (CST 6301) — *Fall 2005–2009, Spring 2006, Fall 2013–2026*
- Cryptography (CST 5335) — *Fall 2007–2008, Spring 2016–2019*
- Computer Networks (CST 5328) — *Fall 2005–2026*
- Parallel Computing (CST 5301) — *Spring 2010, 2012, Fall 2024–2025*

ADVISING

Master's Thesis / Project Advisor

- Shymik Smith, “Robust Pick-and-Place Pipeline for Collaborative Robotic Arms: Toward Autonomous Unmanned Kitchen Systems,” May 2026. (Project)

- Kevin A. Rodriguez Fuentes, "ShuttleFlow: Improving Shuttle Tracking Around Campus," December 2025. (Project Co-advisor)
- Emmanuel White, "Development of an E-Commerce Platform," May 2025. (Project Co-advisor)
- Pamela Baucum, "Development of Framework to Control Bluetooth-Based Robot Car Using an Arduino Microcontroller," May 2020. (Project)
- Jazmyn Saunders, "Design and Development of a Remote Door Locking System Using Arduino," May 2019. (Project)
- Hitesh Pambhar, "An Implementation for Framework for Remote Alarm System Using an Arduino Microcontroller," November 2018. (Project)
- Magdeldin Abdelrahman, "Performance Evaluation of Known Class of TCP using ns-2 Simulation," WSSU, November 2014. (Project)
- Bendhi Eliassaint, "Performance Evaluation of TCP with Various Traffic Patterns using ns-2," WSSU, May 2014. (Project)
- Kristyn Daney, "Implementation of Round Trip Time Simulator with ns-2," WSSU, May 2012. (Project)
- Howard Brown, "A Backoff Scheme for Multicasting in IEEE 802e Wireless Networks," WSSU, May 2012. (Thesis)
- Olasumbo Adeyeye, "Numerical Performance Analysis for Data Aggregation in Wireless Sensor Networks," WSSU, May 2011. (Project)
- Edward Jordan, "Operations of BS with Mobile Robot for Wireless Sensor Networks," WSSU, December 2010. (Thesis)
- Basheer Olusegun Idowu, "Implementation of (N, 1) Steganography Approach for Secret Sharing," WSSU, April 2010. (Project)
- Tarandeep Kaur, "Energy Efficient Deployment and Cluster Formation for Wireless Sensor Networks," WSSU, November 2008. (Thesis)
- Geethanjali Ramaiah, "On Demand Multimedia Service using an Advanced Multicast Technique," WSSU, November 2008. (Project)
- Derrick Hargrove, "An Urgent NAK with Prediction of Lost Packets for Fast Error Recovery in a Reliable Multicast," WSSU, May 2008. (Thesis)
- Kaur Gurpreet, "A New Hybrid Caching Architecture for Cooperative Caching," WSSU, May 2007. (Thesis)
- Wood Kanampiu, "A NAK Suppression Scheme for Group Communications Considering the Spatial Locality of Packet Losses," WSSU, December 2006. (Thesis)

Master's Thesis / Project and Ph.D. Dissertation Committee Member

- Han Ying, "GRPO Reinforcement Learning for Reducing Quantum Circuit Gate Count in the NISQ Era," May 2026. (Korea University, Thesis)
- Yuanjie Li, "Quantum Principles in Concurrency and Quantum Cross Hybrid Fine-Tuning Strategies for LLMs in Multi-Tier IoT Resource Allocation," February 2026. (Korea University, Dissertation)
- Zarin Tasnim Shejuti, "Integrated Postural Assessment and Indoor Localization for Healthcare Workers Using Edge Devices," December 2025. (Thesis)
- Kemai Parker, "Exploring Factors Contributing to Low Academic Performance in Winston-Salem Forsyth County Public Schools (WSFCS)," December 2023. (Project)
- Anthony Smith, "Effectiveness of a Bluetooth RSSI-based Vectorized Algorithm for Mobility Detection within Indoor Space," April 2023. (Thesis)
- Jamal Pete, "Data Mining: A Comparison between Rules from Associative Learning and Finite Inductive Sequences," November 2012. (Thesis)
- Salem Alghamdi, "Secure Payment Transaction Using Two-Factor Authentication," WSSU, May 2011. (Thesis)
- Sultan Almalki, "Credit Card Online Security," WSSU, May 2011. (Thesis)
- Elihah Myers, "FI Algorithm: Achieving Syntactic Pattern Recognition in Finite Inductive Strings," WSSU, April 2010. (Thesis)
- Tina R. Westphal, "Performance Analysis of Wavelet Compression vs. Discrete Cosine Compression in a Medical Imaging Archive Environment," WSSU, April 2010. (Project)
- Shaun Damon Southerland, "The Benefits of Using Pay Per Click Advertising for E-Commerce," WSSU, November 2008. (Project)
- Lamonica Stone, "Data Compression," WSSU, November 2008. (Project)
- Srinivasula Reddy Iragamreddy, "Intrusion Detection Using Apriori Algorithm," December 2007. (Thesis)
- John McDonald, "Topic Maps for Mobile Computing," WSSU, December 2006. (Thesis)
- Dandan Wang, "Searching and Querying Topic Maps in TM4L," WSSU, July 2006. (Thesis)

RESEARCH PUBLICATIONS

Refereed Journals

- [1] **Jinsuk Baek**, Sanhong Lee, and Ju Hong Park, "A Lightweight Closed-Form RGB-D Image Processing Pipeline for Real-Time Yaw Estimation in Resource-Constrained Robotic Systems," *Journal of Real-Time Image Processing (JRTIP)* (Ready to submit).
- [2] Munene W. Kanampiu, and **Jinsuk Baek**, "A Secure and Scalable Internet of Things Smart Campus Network: Design and Configuration," *International Journal of Informatics and Communication Technology* (Submitted, Under review).
- [3] **Jinsuk Baek**, and Minh Jo, "Can Quantum Computing Power Intelligent Systems?" *IEEE IT Professional* (Submitted, Under review).
- [4] YuanJie Li, **Jinsuk Baek**, Shinhoo Kang, and Minh Jo, "PQA: Pure-Quantum Circuit Adaptation for Parameter-Efficient LLM Fine-Tuning," *IEEE Transactions on Neural Networks and Learning Systems* (Submitted, Under review).
- [5] **Jinsuk Baek**, "Workload-Aware Error Mitigation for Consumer-Oriented Quantum Cloud Computing: Practical Evaluation and Insights on NISQ Devices," *IEEE Transactions on Consumer Electronics* (Submitted, Under review).
- [6] YuanJie Li, **Jinsuk Baek**, Shinhoo Kang, and Minh Jo, "Control-Plane-Driven Adaptive Routing for Time-Varying Quantum Repeater Networks," *IEEE Transactions on Network Science and Engineering* (Submitted, Under review).
- [7] **Jinsuk Baek**, and Minh Jo, "No Cloning, No Retransmission: Reliable Multicast Architecture and Open Challenges for the Quantum Internet," *IEEE Network* (Submitted, Under review).
- [8] **Jinsuk Baek**, Sanhong Lee, and Ju Hong Park, "Uncertainty-Aware RGB-D Vision Sensing for Reliable Robotic Sorting," *IEEE Sensors Journal* (Revision Submitted, Under review).
- [9] **Jinsuk Baek**, Sanhong Lee, and Ju Hong Park, "Dual-Arm Synchronization Framework for Smart Kitchen Robot Systems," *IEEE Access* (Accepted).
- [10] **Jinsuk Baek**, Sanhong Lee, and Ju Hong Park, "RL-ACK: A Reinforcement Learning-Based Adaptive ACK Framework for Heterogeneous Smart Kitchen IoT Networks," *IEEE Access*, vol. 14, pp. 100508–100523, July 2026.
- [11] **Jinsuk Baek**, Sanhong Lee, and Ju Hong Park, "Collaborative Multi-Robot Architecture for Smart Kitchens," *IEEE Consumer Electronics Magazine*, pp. 1–8, April 2026.
- [12] YuanJie Li, Kyungmin Lim, **Jinsuk Baek**, and Minh Jo, "QCHFT: Quantum Cross-Hybrid Fine-Tuning for LLMs," *IEEE Transactions on Quantum Engineering*, vol. 7, Article no. 2101021, pp. 1–21, May 2026.
- [13] YuanJie Li, **Jinsuk Baek**, and Minh Jo, "Quantum Principles in Concurrency: Toward Resolving Classical Synchronization Problems," *KSII Transactions on Internet and Information Systems*, vol. 20, no. 5, pp. 2773–2796, May 2026.
- [14] Shiva Dahal, John T. Yi, and **Jinsuk Baek**, "Hardware-Aware Quantum Education: Six Modules Bridging Theory, Hardware, and Software for Next-Generation Quantum Workforce," *KSII Transactions on Internet and Information Systems*, vol. 20, no. 5, pp. 2661–2686, May 2026.
- [15] Cheonshik Kim, Ching-Nung Yang, **Jinsuk Baek**, and Lu Leng, "Survey on Data Hiding Based on Block Truncation Coding," *Applied Sciences*, vol. 11, issue 19, 9209, October 2021.
- [16] **Jinsuk Baek**, Munene W. Kanampiu, and Cheonshik Kim, "A Secure Internet of Things Smart Home Network: Design and Configuration," *Applied Sciences*, vol. 11, issue 14, 6280, July 2021.
- [17] Paul S. Fisher, Jimmy James II, **Jinsuk Baek**, and Cheonshik Kim, "Mining Intelligent Solution to Compensate Missing Data Context of Medical IoT Devices," *Personal and Ubiquitous Computing*, vol. 22, issue 1, pp. 219–224, February 2018.
- [18] Doowon Kim, **Jinsuk Baek**, Paul S. Fisher, and Sangchul Kim, "fFTP: A Fast File Transfer Protocol for Home N-Screen Platform," *Personal and Ubiquitous Computing*, vol. 22, issue 1, pp. 143–152, February 2018.
- [19] Paul S. Fisher, Nawaf Aljohani, and **Jinsuk Baek**, "Generation of Finite Inductive, Pseudo Random, Binary Sequences," *Journal of Information Processing Systems*, vol. 13, issue 6, pp. 1554–1574, December 2017.
- [20] **Jinsuk Baek**, Cheonshik Kim, "An Energy-Efficient Video Transport Protocol for Personal Cloud-Based Computing," *Journal of Real-Time Image Processing (JRTIP)*, vol. 12, issue 2, pp. 303–310, June 2016.
- [21] **Jinsuk Baek**, Min Gyung Kwak, and Paul S. Fisher, "A Smart Backoff Scheme for Multicast Flows in Wireless LAN," *The Smart Computing Review (Smart CR)*, vol. 5, issue 3, pp. 159–167, June 2015. (Invited Paper)
- [22] Paul S. Fisher, **Jinsuk Baek**, and Howard Fisher, "APPMPS: An Asynchronous, Parallel, Plastic, Multi-Processing System," *IEEE*

Systems Journal, vol. 9, issue 2, pp. 382–392, June 2015.

- [23] Yong woon Ahn, Albert M. K. Cheng, **Jinsuk Baek**, Minh Jo, and Hsiao-Hwa Chen, “An Auto-Scaling Mechanism for Virtual Resources to Support Mobile, Pervasive, Real-time, Healthcare Applications in Cloud Computing,” *IEEE Network*, vol. 27, issue 5, pp. 62–68, October 2013.
- [24] Mohammad M. Fuad, Dezbani Deb, and **Jinsuk Baek**, “Static Analysis, Code Transformation and Runtime Profiling for Self-healing,” *Journal of Computers (JCP)*, vol. 8, issue 5, pp. 1127–1135, May 2013.
- [25] **Jinsuk Baek**, Min Gyung Kwak, and Paul S. Fisher, “A Hybrid Upstream Bandwidth Allocation Method for Multimedia Communications in EPONs,” *IEEE Transactions on Smart Processing and Computing*, vol. 1, issue 1, pp. 27–33, August 2012. (Invited Paper)
- [26] **Jinsuk Baek**, Paul S. Fisher, Minh Jo, and Hsiao-Hwa Chen, “A Reliable Overlay Video Transport Protocol for Multicast Agents in Wireless Mesh Networks,” *Wiley International Journal of Communication Systems*, vol. 25, issue 5, pp. 553–570, May 2012.
- [27] **Jinsuk Baek**, Doowon Kim, Paul S. Fisher, and Minh Jo, “An Adaptive Primary Path Switching Scheme for Seamless mSCTP Handover,” *The Smart Computing Review (Smart CR)*, vol. 1, issue 2, pp. 150–160, December 2011. (Invited Paper)
- [28] Yong woon Ahn, **Jinsuk Baek**, Albert M. K. Cheng, Paul S. Fisher, and Minh Jo, “A Fair Transmission Opportunity by Detecting and Punishing the Malicious Wireless Stations in IEEE 802.11e EDCA Network,” *IEEE Systems Journal*, vol. 5, issue 4, pp. 486–494, December 2011.
- [29] **Jinsuk Baek**, Jan Fischer, Hsiao-Hwa Chen, and Minh Jo, “On a Moving Direction Pattern Based MAP Selection Model for HMIPv6 Networks,” *Elsevier Journal of Computer Communications*, vol. 34, issue 2, pp. 150–158, February 2011.
- [30] **Jinsuk Baek**, Sun Kyong An, and Paul S. Fisher, “Dynamic Cluster Header Selection and Conditional Re-clustering for Energy Efficient Wireless Sensor Networks,” *IEEE Transactions on Consumer Electronics*, vol. 56, issue 4, pp. 2249–2257, November 2010.
- [31] **Jinsuk Baek**, Paul S. Fisher, Minh Jo, and Hsiao-Hwa Chen, “A Lightweight SCTP for Partial Reliable Overlay Video Multicast Service for Mobile Terminals,” *IEEE Transactions on Multimedia*, vol. 12, issue 7, pp. 754–766, November 2010.
- [32] Tarandeep Kaur, and **Jinsuk Baek**, “A Strategic Deployment and Cluster-Header Selection for Wireless Sensor Networks,” *IEEE Transactions on Consumer Electronics*, vol. 55, issue 4, pp. 1890–1897, November 2009.
- [33] Young woon Ahn, Albert Cheng, **Jinsuk Baek**, and Paul S. Fisher, “A Multiplayer, Real Time, Game Protocol Architecture to Reduce Network Latency,” *IEEE Transactions on Consumer Electronics*, vol. 55, issue 4, pp. 1883–1889, November 2009.
- [34] **Jinsuk Baek**, Kaur Gurpreet, and Junghoon Yang, “A New Hybrid Caching Architecture for Cooperative Caching,” *Journal of Ubiquitous Convergence Technology*, vol. 2, no. 1, pp. 1–11, August 2008.
- [35] **Jinsuk Baek** and Jehan-François Pâris, “A Scalable Recovery Tree Construction Considering Packet Losses Correlation for Reliable Multicast,” *KSII Transactions on Internet and Information Systems*, vol. 2, no. 2, pp. 82–102, April 2008.
- [36] **Jinsuk Baek**, Paul S. Fisher, and Mingyung Kwak, “FI-Based Local Group Key Generation/Distribution for Mobile Multicast in a Hierarchical Mobile IPv6 Network,” *KSII Transactions on Internet and Information Systems*, vol. 2, no. 1, pp. 5–22, February 2008.
- [37] **Jinsuk Baek**, Cheonshik Kim, and YouSik Hong, “Packet Loss Patterns Adaptive Feedback Scheduling for Reliable Multicast,” *Journal of Ubiquitous Convergence Technology*, vol. 1, no. 1, pp. 28–34, November 2007.
- [38] **Jinsuk Baek** and Wood Kanampiu, “A NAK Suppression Scheme for Group Communications Considering the Spatial Locality of Packet Losses,” *International Journal of Computer Science and Network Security*, vol. 6, no. 10, pp. 158–167, October 2006.
- [39] **Jinsuk Baek** and Eunjung Lee, “A Dynamic Authentication, Authorization, and Accounting (AAA) Resolution for Hierarchical Mobile IPv6 Network,” *International Journal of Computer Science and Network Security*, vol. 6, no. 7B, pp. 170–179, July 2006.
- [40] **Jinsuk Baek** and Eunjung Lee, “A Dual Mode Buffer for Reliable Multicast in Mobile IP Networks,” *International Journal of Computer Science and Network Security*, vol. 6, no. 5B, pp. 1–8, May 2006.
- [41] **Jinsuk Baek** and Jehan-François Pâris, “A Heuristic Buffer Management and Retransmission Control Scheme for Tree-Based Reliable Multicast,” *ETRI Journal*, vol. 27, no. 1, pp. 1–12, February 2005.

Refereed Conferences with Proceedings

- [1] **Jinsuk Baek**, and Munene W. Kanampiu, “Criticality-Aware Hybrid Networking for Scalable and Resilient Robotic Smart

Kitchens,” ICACT 2027 (under review).

- [2] **Jinsuk Baek**, “Reliable Short Control Messaging via Multi-Route Quantum Superdense Coding for Future Consumer Networks,” IEEE CCNC 2027, January 2027 (under review).
- [3] Nisha Somai, Dahyeong Jeong, **Jinsuk Baek**, and Minho Jo, “Hyperbolic Large Concept Models: Geometry-Aware Hierarchical Reasoning Beyond Euclidean Space,” NeurIPS 2026, December 2026 (under review).
- [4] Kyungmin Lim, **Jinsuk Baek**, Munene W. Kanampiu, and Minho Jo, “Auxiliary-Branch Complexity in Mamba for Efficient LLMs,” INTCOM 2026, July 2026.
- [5] Kyungmin Lim, **Jinsuk Baek**, and Minho Jo, “A Teleportation-Based Quantum Repeater with Multi-Route Redundancy and Stabilizer Verification,” AIRC 2026, April 2026.
- [6] Munene W. Kanampiu, and **Jinsuk Baek**, “A Cooperative Retransmission Control Scheme for CoAP in Constrained IoT Networks,” ICACT 2026, February 2026. (Outstanding Paper Award)
- [7] Sunghwan Kim, I Gede Pasek Suta Wijaya, **Jinsuk Baek**, and Minho Jo, “Bayesian Deep Learning Approach for Traffic Accident Severity Prediction with Uncertainty Estimation,” ICONI 2025, December 2025.
- [8] Shymik Smith, Munene W. Kanampiu, and **Jinsuk Baek**, “A Robust Pick-and-Place Pipeline for Collaborative Robotic Arms: Toward Autonomous Unmanned Kitchen Systems,” CSCI 2025, December 2025.
- [9] Youngwon Yu, Kyungmin Lim, Taras Maksmyuk, **Jinsuk Baek**, Vladimir Simic, and Minho Jo, “RAG-Based Generative Large Language Model for University Regulations,” APIC-IST 2025, July 2025.
- [10] Hitesh Pambhar, and **Jinsuk Baek**, “Enhancing MQTT for Targeted Communication in Automated Smart Homes,” CSCI 2024, December 2024.
- [11] Kemai Parker, Munene W. Kanampiu, and **Jinsuk Baek**, “Exploring Factors Contributing to Low Academic Performance in Winston-Salem Forsyth County Public Schools (WS/FCS),” IACIS Annual Conference 2024, October 2024.
- [12] Ahmad Hijaz, Ibrahim Yakubu, and **Jinsuk Baek**, “Integrating Secure Quantum Digital Signature into Quantum Communications,” ICEQT 2024, July 2024.
- [13] Makayla Morgan, Ahmad Hijaz, and **Jinsuk Baek**, “WSSU Quantum Mini Tennis: Learning Quantum Gates and State Vectors Through Gaming,” ADMI 2024, April 2024.
- [14] Teo Kim, **Jinsuk Baek**, and John Yi, “Quantum Authentication Protocol for Secure Quantum Superdense Coding,” IEEE SoutheastCon 2024, March 2024.
- [15] John Yi, **Jinsuk Baek**, John Merle, Xiuping Tao, and Kasra Sardashiti, “Preparing a Diverse Quantum Workforce in Carolinas through Hands-on Education and Training Experiences,” APS March Meeting 2024, March 2024.
- [16] Jaylin Butts, Emmanuel White, **Jinsuk Baek**, “A Quantum Teleportation Protocol Secured by a Blockchain Technology,” ICEQT 2023, pp. 1665–1669, July 2023.
- [17] **Jinsuk Baek**, and Munene W. Kanampiu, “A MQTT-based Smart Home Monitoring System: Design and Implementation,” International Conference on Wireless Sensor Networks, Ubiquitous Computing and Applications, December 2022.
- [18] **Jinsuk Baek**, and Munene W. Kanampiu, “A MQTT-based Soil Moisture Level Notification System for a Smart Farm,” IEEE ICPEA 2021, October 2021.
- [19] **Jinsuk Baek**, and Munene W. Kanampiu, “A Strategic Sensor Placement for a Smart Farm Water Sprinkler System: A Computational Model,” IEEE ICACT 2021, February 2021.
- [20] **Jinsuk Baek**, Munene W. Kanampiu, and Byeong Hyun Ko, “A Set Notation Configuration CoAP Scheme for Internet of Things,” IEEE ICACT 2020, February 2020. (Outstanding Paper Award)
- [21] **Jinsuk Baek**, Munene W. Kanampiu, “An Evaluation of How Much HTTP/1.1 Outperforms HTTP/1.0,” ICSTEM 2019, pp. 1–3, March 2019.
- [22] Hitesh Pambhar, and **Jinsuk Baek**, “An Implementation for Framework for Remote Alarm System Using an Arduino Microcontroller,” ICRAET 2018, pp. 1–3, October 2018.
- [23] Geethanjali Ramaiah, **Jinsuk Baek**, “Client-Based, On-Demand Multimedia Service Using an Advanced Multicast Transmission Technique,” SICEAS, pp. 123–125, December 2017.
- [24] **Jinsuk Baek**, Paul S. Fisher, “Packet Damage-Resistant Analysis Using a Data Mining Mechanism in Wireless Sensor Networks,” IEEE IE 2017, pp. 176–179, August 2017.

- [25] **Jinsuk Baek**, Paul S. Fisher, "A Simple Broadcast-Based Service Discovery for Home-Based Lighting Control Network," SICEAS, pp. 455–457, February 2017.
- [26] Jamal Pete, Paul S. Fisher, and **Jinsuk Baek**, "Reliable Decision Making: A Comparison between Rules from Associative Learning and Finite Inductive Sequences," APARM 2016, pp. 461–468, August 2016.
- [27] Min Gyung Kwak, **Jinsuk Baek**, and Paul S. Fisher, "Retransmission Control Scheme for RDM Protocol for Lightning Control Networks," IEEE ICACT 2016, pp. 86–90, February 2016. (Outstanding Paper Award)
- [28] Min Gyung Kwak, **Jinsuk Baek**, and Paul S. Fisher, "A Degree Constrained Spanning Tree-Based Device Discovery Scheme for Lightning Control Networks," IEEE IoT 2015, pp. 1–2, October 2015.
- [29] Cheonshik Kim, **Jinsuk Baek**, and Paul S. Fisher, "Lossless Data Hiding for Binary Document Images Using n-Pairs Pattern," ICISC 2014, pp. 317–327, December 2014.
- [30] Paul S. Fisher, Min Gyung Kwak, Eunjung Lee, and **Jinsuk Baek**, "A Signature Scheme for Digital Imagery," ICISA 2014, pp. 115–118, May 2014.
- [31] Doowon Kim, **Jinsuk Baek**, and Paul S. Fisher, "Adaptive Video Streaming over HTTP," ACMSE 2014, March 2014. (poster session)
- [32] Doowon Kim, Chaitanya Achan, **Jinsuk Baek**, and Paul S. Fisher, "Implementation of Framework to Identify Potential Phishing Websites," IEEE ISI 2013, pp. 268, June 2013. (poster session)
- [33] Howard Brown, Min Gyung Kwak, **Jinsuk Baek**, and Paul S. Fisher, "Impact of Multicast Flow for Performance of IEEE 802.11e in Wireless LAN," IEEE ICACT 2013, pp. 768–773, January 2013.
- [34] Hyunju Jo, Sangchul Kim, **Jinsuk Baek**, and Paul S. Fisher, "Motion Recognition with Smart Phone Embedded 3-Axis Accelerometer Sensor," IEEE SMC 2012, pp. 919–924, October 2012.
- [35] Wood Kanampiu, Justin Zhan, and **Jinsuk Baek**, "A Secure Group Collaboration Protocol for Non-Verbal Human Social Signals Featuring Deception Detection," IEEE PASSAT 2012, pp. 813–818, September 2012.
- [36] Illyoung Choi, **Jinsuk Baek**, Sangchul Kim, and Paul S. Fisher, "A New File System Specialized for N-Screen Platforms," IEEE ICACT 2012, pp. 231–234, February 2012. (Outstanding Paper Award)
- [37] Mohammad M. Fuad, Dezbani Deb, and **Jinsuk Baek**, "Self-Healing by means of Runtime Execution Profiling," ICCIT 2011, pp. 202–207, December 2011.
- [38] Yong woon Ahn, Chaitanya Belwal, Albert M. K. Cheng, and **Jinsuk Baek**, "Improving QoS for ECG Data Transmission with Enhanced Admission Control in EDCA-Based WLANs," IEEE GLOBECOM 2011, pp. 1–5, December 2011.
- [39] Edward Jordan, **Jinsuk Baek**, and Wood Kanampiu, "Impact of Mobile Sink for Wireless Sensor Network," ACM SE 2011, pp. 328–329, March 2011. (poster session)
- [40] Elijah Myers, Paul S. Fisher, Keith Irwin, **Jinsuk Baek**, and Joao Setubal, "The Complexity and Application of Syntactic Pattern Recognition Using Finite Inductive Strings," IAPR PRIB 2010, LNCS 5780, pp. 125–136, September 2010.
- [41] **Jinsuk Baek**, Paul S. Fisher, and Minho Jo, "An Enhancement of mSCTP Handover with an Adaptive Primary Path Switching Scheme," IEEE VTC, September 2010.
- [42] Paul S. Fisher, John O. Adeyeye, **Jinsuk Baek**, and Joao Carlos Setubal, "Finite Inductive Sequences, Kolmogorov Complexity with Application to Genome Sequences," BCBGC-10, pp. 78–83, July 2010.
- [43] **Jinsuk Baek**, Cheonshik Kim, Paul S. Fisher, Hongyang Chao, "(N, 1) Steganography Approach for Secret Sharing with Digital Images," IEEE WCNIS 2010, pp. 325–329, June 2010.
- [44] Yong woon Ahn, Albert M. K. Cheng, **Jinsuk Baek**, and Paul Fisher, "Detection and Punishment of Malicious Wireless Stations in IEEE 802.11e EDCA Network," IEEE Sarnoff 2010, pp. 1–5, April 2010.
- [45] Paul S. Fisher, **Jinsuk Baek**, Minho Jo, "Program Readability: A Proposed Software Metric," ICONI & APIC-IST 2009, December 2009.
- [46] Paul S. Fisher, Howard Fisher, **Jinsuk Baek**, and Cleopas Angaye, "Syntactic Pattern Recognition Using Finite Inductive Strings," IAPR PRIB 2009, LNCS 5780, pp. 89–101, September 2009.
- [47] **Jinsuk Baek**, Min Gyung Kwak, Paul S. Fisher, and Elva J. Jones, "A Web Object Management Policy for Cooperative Hybrid Caching Architecture," IEEE ICCSIT 2009, pp. 61–65, August 2009.
- [48] **Jinsuk Baek**, Sun Kyong An, Paul S. Fisher, and Elva J. Jones, "Dynamic Cluster Header Selection with Self-incentive for Wireless

Sensor Networks,” IEEE Sarnoff 2009, pp. 1–5, March–April 2009.

- [49] **Jinsuk Baek**, Paul S. Fisher, and Mingyung Kwak, “FI-based File Access Predictor,” ACM SE 2009, pp. 1–4, March 2009.
- [50] **Jinsuk Baek**, Derrick Hargrove, Paul S. Fisher, and Cheonshik Kim, “An Urgent NAK with Prediction of Lost Packets for Fast Error Recovery in a Reliable Multicast,” IWUCT 2008, pp. 113–120, August 2008.
- [51] **Jinsuk Baek**, Paul S. Fisher, Mingyung Kwak, and Hyungoo Han, “Secure Scalable and Reliable Multicast with FI-Based Local Group Key,” ICOMP 2008, pp. 188–194, July 2008.
- [52] **Jinsuk Baek**, Paul S. Fisher, Yuri Kim, Mingyung Kwak, and Hyungoo Han, “An Object Discarding Policy with FI Systems for Cooperative Hybrid Caching Architecture,” ACM SE 2008, pp. 411–416, March 2008.
- [53] YouSik Hong, Cheonshik Kim, and **Jinsuk Baek**, “A Study on the Traffic Accident Examination Using Artificial Intelligence,” IUC/IUW 2007, pp. 41–46, July 2007.
- [54] **Jinsuk Baek** and Wood Kanampiu, “An Admission Control Scheme for Repair Server in a Tree-Based Reliable Multicast,” ACM SE 2007, pp. 191–196, March 2007.
- [55] Eunjung Lee, **Jinsuk Baek**, and Stephen S.-H. Huang, “A Dynamic Mobility Management Scheme for VoIP Services,” ITNG 2006, pp. 340–345, April 2006.
- [56] **Jinsuk Baek** and Jehan-François Pâris, “A Tree-Based Reliable Multicast Scheme Exploiting the Temporal Locality of Transmission Errors,” IEEE IPCCC 2005, pp. 275–282, April 2005.
- [57] **Jinsuk Baek** and Jehan-François Pâris, “An Efficient Retransmission Control Scheme for Tree-Based Reliable Multicast,” SPECTS 2004, pp. 145–152, July 2004.
- [58] **Jinsuk Baek** and Jehan-François Pâris, “A Heuristic Buffer Management Scheme for Tree-Based Reliable Multicast,” IEEE ISCC 2004, pp. 1123–1128, June–July 2004.
- [59] **Jinsuk Baek** and Jehan-François Pâris, “An Efficient Buffer Management Scheme for Tree-Based Reliable Multicast Using Infrequent Acknowledgements,” IEEE IPCCC 2004, pp. 13–20, April 2004.
- [60] **Jinsuk Baek** and Eunjung Lee, “An Improved Logical Tree Construction Scheme for Reliable Multicast,” IFIP ICTS 2003, pp. 110–121, October 2003.
- [61] **Jinsuk Baek**, “A New Configuration Scheme for ACK Tree in Reliable Multicast Protocols,” ICC 2002, pp. 972–987, August 2002.
- [62] **Jinsuk Baek**, “A Hybrid Configuration of ACK Tree for Multicast Protocol,” SPECTS 2002, pp. 852–856, July 2002.
- [63] **Jinsuk Baek** and Sangchul Kim, “An Efficient MF (Model-Failure) Object Coding Method for Object-Oriented Coder,” 24th KISS Fall Conference, pp. 131–134, November 1997.
- [64] Heekyung Kim, **Jinsuk Baek**, and Sangchul Kim, “CAI System for C and UNIX Using a Hypermedia Technology,” 22nd KISS Spring Conference, pp. 1063–1066, April 1996.

Dissertation, Thesis, and Miscellaneous

- [1] Jinsuk Baek, “Scalable Approaches for Tree-Based Reliable Multicast,” Ph.D. Dissertation, University of Houston, December 2004.
- [2] Jinsuk Baek, “An Efficient Compression Method for Broadcasting Protocol in Low Bit Transmission Rate,” Bronze Prize for Good Research Paper, 1st SAMSUNG Electronics New Technology Research Contest, December 2000.
- [3] Jinsuk Baek, “An Efficient Parameter Coding Method for Object-Oriented Coder (MPEG-4),” Master's Thesis, Hankuk University of Foreign Studies, February 1998.

GRANT PROPOSALS

- [1] “Winston-Salem Advanced Quantum Training, Education & Readiness (WAQTER) Program,” U.S. Army Research Office, LPS Qubit Collaboratory (LQC). (Co-PI, Submitted Apr 30, 2026, \$2,000,000)
- [2] “AI-Enabled Smart Manufacturing Platform for Real-Time 3D Electrode Microstructure Analysis in Lithium Metal Battery Manufacturing,” U.S. DoE Genesis Mission, Phase I. (PI, Submitted Apr 28, 2026, \$388,583; WSSU portion \$100,493)
- [3] “A Novel Low-Cost Robotic Assistant Integrating AI, 3D Printing, Embedded Microcomputer Programming, Sensor-Based Control,

- and Security for STEM Education,” 2025 WSSU Catalyst Program for Research Funding. (Co-PI, Funded \$7,500)
- [4] “Acquisition of an Advanced Collaborative Robotic Arm for Interdisciplinary Research and Teaching at WSSU,” 2025 WSSU Equipment Grant Program. (PI, \$13,868, Not Funded)
 - [5] “Southeast Quantum Innovation Engine (SEQIE),” National Science Foundation. (PI, Submitted Apr 15, 2025, \$1,280,186, Not Funded)
 - [6] “Cooking Robot System for Everybody,” Korea Ministry of Agriculture, Food and Rural Affairs. (PI, 2024–2027, **Funded \$3,000,000; WSSU portion \$180,000**)
 - [7] “Innovate Education and Research Training for Quantum Computing (QC): Empowering Underrepresented Minority Students (UMS),” U.S. Department of Education. (PI, Submitted May 28, 2023, \$899,997, Not Funded)
 - [8] “Expanding Quantum Research and Education at Winston-Salem State University with Research on Hybrid Microwave-Optical Quantum Devices,” National Science Foundation. (Senior Personnel, 2023–2028, **Funded \$5,000,000**)
 - [9] “Training a Diverse Quantum Workforce in Carolinas through Establishing the Winston-Salem Quantum Education Collaboratory (WS-QEC),” Department of Energy. (Co-PI, 2023–2026, **Funded \$2,253,963**)
 - [10] “Adjusting Level of Simultaneous Updates Using a Set Configuration CoAP Scheme for Internet of Things,” IBM Center for The Business of Government Research Stipend Program. (PI, Submitted Mar 25, 2020, \$20,000, Not Funded)
 - [11] “Middleware API Development for Heterogeneous Device Management for IoT,” National Science Foundation. (Co-PI as invited scholar, Submitted Nov 2017, Not Funded)
 - [12] “Asynchronous, Parallel, Plastic, Multi-Processing System (APPMPS): The Memory System,” Department of Defense. (Co-PI, Submitted Jan 2015, \$593,661, Not Funded)
 - [13] “Development of an Extensible File Synchronization System across Multiple, Heterogeneous Computing Devices,” Department of Defense. (PI, Submitted Jan 2015, \$266,070, Not Funded)
 - [14] “A Lightweight Transport Protocol to Support Reliable N-Screen Services,” WSSU Research Initiation Program (RIP) 2013. (PI, Funded \$10,000)
 - [15] “Mobile Wireless Computer and Communication Systems and IT-BT Convergence for Education,” Department of Defense. (PI, Submitted Sep 2010, \$90,658, Not Funded)
 - [16] “Decision Support Model Based on Information Acquired by Dynamically Optimized Sensors-to-Commander Communications,” Department of Defense. (Co-PI, Submitted Oct 2010, \$1,993,300, **Recommended but not Funded**)
 - [17] “Design and Implementation of (N, 1) Secret Sharing Approach Based on Steganography with Gray Digital Images,” IBM Center for The Business of Government. (PI, Submitted Aug 2010, \$20,000, Not Funded)
 - [18] “Operations of BS with Mobile Robot for Wireless Sensor Networks,” 2010–2011 NC Space Graduate Fellowship. (Research Advisor, **Funded \$6,000**)
 - [19] “Parallel Computing Infrastructure to Advance Research and Education Capabilities,” National Science Foundation. (Co-PI, Submitted Aug 2009, \$306,272 for WSSU / ~\$600,000 total with UNCG, Not Funded)
 - [20] “A Wireless Protocol for Prediction of Next Service for an All-Mobile Environment,” Army Research Office. (Co-PI, Submitted Mar 2008, \$368,215, Not Funded)
 - [21] “Ad Hoc Network, Security and a Mobile Protocol,” National Science Foundation. (PI, Submitted Mar 2008, \$448,386, Not Funded)
 - [22] “Smart, Mobile, Ad Hoc, Sensor Networks for Utilization by the Modern War-Fighter Applied to Surveillance in Battlefield Space,” Army Research Office. (Co-PI, Submitted Jan 2008, \$799,904, **Recommended but not Funded**)
 - [23] “WSSU DURIP: WiFi, IEDs Video Systems and Education,” Department of Defense. (Co-PI, Submitted Aug 18, 2007, \$127,136.23, Not Funded)
 - [24] “A Dynamic Admission Control Scheme for Reliable Multicast,” 2007–2008 NC Space Graduate Fellowship. (Research Advisor, Funded \$5,000)
 - [25] “Performance Evaluation of Novel Hybrid Cache Architecture for Cooperative Web Caching,” WSSU Research Initiation Program (RIP) 2007. (PI, **Funded \$10,000**)
 - [26] “A NAK Suppression Scheme for Group Communications Considering the Spatial Locality of Packet Losses,” 2006–2007 NC Space Graduate Fellowship. (Research Advisor, **Funded \$5,000**)

- [27] “Seamless Multicast for Mobile Hosts in Mobile IPv6 Networks,” WSSU Research Initiation Program (RIP) 2006. (PI, Submitted Feb 2006, Not Funded)
- [28] “Tablet PCs in Teaching and Learning,” 2006 HP Equipment Support Program. (Co-PI, Submitted Feb 2006, Not Funded)

PROFESSIONAL ACTIVITIES

Memberships & Leadership

- Member: IEEE, KSII, and IEEK
- Director: IEEK Computer Society

Journal Editorial Roles

- Guest Editor, Journal of Real-Time Image Processing — Special Issue on Real-Time Data Hiding and Visual Cryptography (2018)
- Guest Editor, International Journal of Distributed Sensor Networks — Special Issue on Data Mining Techniques based on Wireless Sensor Networks (2017)
- Editorial Board, Security and Communication Networks (2017 – Present)
- Guest Editor, International Journal of Security and Communication Networks — Special Issue on Emerging Security and Privacy Influencing Enterprise BYOD and Future Mobile Network (2017)
- Guest Editor, Springer Journal of Real Time Processing — Special Issue on Real-Time Data Hiding and Visual Cryptography (2015)
- Editorial Board, IEIE Transactions on Smart Processing and Computing (IEEK SPC) (2012 – Present)
- Editorial Board, International Journal of Computer Science and Network Security (IJSNS) (2006 – Present)
- Editorial Board, The Smart Computing Review (Smart CR) (2011 – 2024)
- Editorial Board, KSII Transactions on Internet and Information Systems (TIIS) (2007 – 2024)

Ph.D. Dissertation External Committee

- YuanJie Li, “Quantum Principles in Concurrency and Quantum Cross Hybrid Fine-Tuning Strategies for LLMs in Multi-Tier IoT Resource Allocation,” Ph.D. Dissertation, Korea University, December 2025.

Conference Technical Program Committee

APIC-IST 2025, SICBS 2017, BigComp 2015, AICT 2015, 3PGCIC 2015, IEEE IPCCC 2014, UMAS 2012, UMAS 2013, IEEE ICCCN 2011, IEEE VTC Fall 2010, ICONI 2009, IEEE ITNG 2007, ACM SE 2007, ACM SE 2008, IUC/IUW 2007, IUC/IUW 2008, ICUCT 2007

Journal Reviewer

- Discover Internet of Things Journal — 2026 – Present
- Agriculture — 2022 – Present
- Electronics — 2022 – Present
- Sensors — 2021 – Present
- IEEE Access — 2020 – Present
- Springer Mobile Networks and Applications — 2017 – Present
- Springer Personal Ubiquitous Computing — 2017 – Present
- Hindawi Security and Communication Networks — 2017 – Present
- Springer Journal of Real Time Processing — 2015 – Present
- IEEE Wireless Communications — 2014 – Present
- Journal of Applied Mathematics — 2013 – Present
- Multimedia Tools and Applications — 2013 – Present
- Hindawi International Journal of Distributed Sensor Networks — 2013 – Present
- EURASIP Journal of Wireless Communications and Networking — 2013 – Present
- IEEE Transactions on Circuits and Systems for Video Technology — 2012 – Present
- IEEE Systems Journal — 2012 – Present
- Springer Cluster Computing — 2012 – Present
- Inderscience Intl. Journal of Intelligent Information and Database Systems — 2011 – Present

- IEEE Transactions on Broadcasting — *2010 – Present*
- Elsevier Computer Networks — *2010 – Present*
- Wiley International Journal of Communication Systems — *2010 – Present*
- IEEE Transactions on Mobile Computing — *2009 – Present*
- Hindawi Journal of Computer Systems, Network, and Communications — *2009 – Present*
- Wiley Wireless Communications and Mobile Computing — *2008 – Present*
- Journal of Supercomputing — *2008 – Present*
- IEEK Journal — *2008 – Present*
- KSII Transactions on Internet and Information Systems (TIIS) — *2007 – Present*
- Journal of Ubiquitous Convergence Technology — *2007 – Present*

Conference Reviewer

SICBS 2017, DemSet 2012–2014, ISA 2012, ACM ICUIMC 2011, CCCT 2011, IEEE CHINACOM 2009, IEEE GLOBECOM 2004, IEEE IPCCC 2004 and 2005

UNIVERSITY & DEPARTMENTAL SERVICE

Department Level — Department of Computer Science, WSSU

- CS Faculty Search Committee Chair — *Spring 2024*
- HBCU Smart Cities Challenge Advisor for Team WSSU — *Fall 2022*
- CS Faculty Representative for WSSU Open House — *Fall 2022 – Spring 2026*
- CS Post-Tenure Review Committee Chair — *Spring 2017, Fall 2022*
- CS Faculty Reappointment Committee Chair — *Spring 2022*
- CS Faculty Meeting Recorder — *Fall 2007–Spring 2008, Fall 2020–Spring 2021*
- CS Faculty Search Committee — *Spring 2010, Spring 2023*
- Recruitment & Retention Committee — *Fall 2008 – Spring 2010*
- Department Enrichment Committee — *Fall 2007 – Present*
- Graduate Admission Committee — *Fall 2006 – Present*
- Undergraduate Program Curriculum/Assessment Committee — *Fall 2006 – Spring 2010*
- Graduate Program Curriculum/Assessment Committee — *Fall 2006 – Present*
- International Initiative for Foreign Students, South Korea — *Fall 2005 – Present*

University Level — Winston-Salem State University

- WSSU Professional Development Committee — *Fall 2017 – Present*
- Recruited 17 International Students from South Korea — *Fall 2006 – Present*
- WSSU Freshman General Education Advisor — *Fall 2012 – Present*
- WSSU IAPP India Planning Task Force — *Spring 2010 – Present*
- WSSU Globalization Task Force — *Fall 2009 – Present*
- WSSU International Program Committee — *Fall 2009 – Present*
- WSSU Faculty Senate — *Fall 2006 – Present*
- WSSU Academic Standards and Curriculum Committee — *Fall 2008 – Present*
- WSSU Constitution Committee — *Fall 2008 – Spring 2010*
- WSSU Graduate Council — *Fall 2007 – Spring 2009*
- Search Committee for International Programs Advisor — *Fall 2008*